

Project 4 – Enterprise OSPF and ROAS Network

Objective

Design and implement an enterprise network connecting a Head Office and two Branch Offices using OSPF dynamic routing. Each site contains multiple VLANs configured using Router-on-a-Stick (ROAS). A centralized DHCP server at the Head Office provides IP addresses to all VLANs across the network using DHCP relay.

Network Components

Routers

- 3 Cisco Routers
 - R1 (Head Office)
 - R2 (Branch A)
 - R3 (Branch B)

Switches

- 3 Layer 2 Switches
- One switch at each office.

End Devices

- 12 PCs
- Two VLANs per office.

Technologies Used

- VLAN Configuration
- Access Ports
- Trunk Links
- Router-on-a-Stick (ROAS)
- OSPF Dynamic Routing
- DHCP Server
- DHCP Relay (IP Helper Address)

- Static IP Addressing for WAN Links
- Inter-VLAN Routing
- End-to-End Network Connectivity

VLAN Structure

Head Office

- VLAN 10 – Staff
- VLAN 20 – Management

Branch A

- VLAN 10
- VLAN 20

Branch B

- VLAN 10
- VLAN 20

Network Design

- R1 acts as the Head Office router.
- R2 and R3 represent branch office routers.
- Router-to-router connections use point-to-point WAN links.
- Each office contains two VLANs.
- ROAS provides inter-VLAN communication.
- OSPF exchanges routing information between all routers.
- R1 operates as the centralized DHCP server.
- R2 and R3 forward DHCP requests using IP Helper Address.

Configuration Tasks

Switch Configuration

- Create VLANs.
- Assign access ports.
- Configure trunk ports connected to routers.

Router Configuration

- Configure ROAS sub-interfaces.
- Assign gateway IP addresses.
- Configure OSPF.
- Configure DHCP pools.
- Configure DHCP relay on branch routers.

End Devices

- Configure DHCP mode.
- Verify automatic IP assignment.

Testing and Verification

The following tests were successfully completed:

VLAN Testing

- Devices communicate within their VLAN.

Inter-VLAN Routing

- VLAN 10 and VLAN 20 communicate through ROAS.

OSPF Verification

- All routers learn remote networks dynamically.

DHCP Verification

- Clients receive IP addresses automatically.
- Branch office clients receive addresses through DHCP relay.

End-to-End Connectivity

- Head Office devices can communicate with Branch A.
- Head Office devices can communicate with Branch B.
- Branch A and Branch B devices can communicate with each other.

Skills Learned

- Enterprise network design.
- VLAN implementation.
- Router-on-a-Stick (ROAS).
- OSPF dynamic routing.
- DHCP server configuration.
- DHCP relay using IP Helper Address.
- Inter-VLAN routing.
- WAN connectivity.
- Network troubleshooting and verification.

Result

Successfully designed and implemented a multi-site enterprise network using VLANs, ROAS, OSPF, and centralized DHCP services. Dynamic routing and DHCP relay enabled seamless communication and automatic IP address allocation across all office locations.

Conclusion

This project demonstrates the implementation of a scalable enterprise network integrating Layer 2 and Layer 3 technologies. By combining VLAN segmentation, Router-on-a-Stick, OSPF dynamic routing, centralized DHCP, and DHCP relay, the network provides efficient communication between multiple office locations while simplifying network management and addressing.